

Bidirectional PDA

An “unrestrictive” PDA is defined by its ability to transmit both pressure and flow across the ductus, resulting in similar pressures in the pulmonary artery and aorta. Directionality is then determined by the PVR/SVR relationship throughout the cardiac cycle. As the duct begins to constrict, pressure transmission becomes progressively limited—this is what we refer to as “restrictiveness.” Importantly, unrestrictive is a physiologic and Doppler-based concept rather than a simple size criterion. Typically, low flow velocities (<2 m/s) are observed in both systole and diastole, reflecting the free transmission of pressure. However, low velocities can also be seen when PVR approaches SVR, even if the duct has started to narrow and behave restrictively. This means that velocity data must always be interpreted in the context of the overall physiology, as ductal size alone does not fully capture its hemodynamic significance.

1. Ductal Diameter

- Usually ≥ 2.5 – 3 mm in term infants (or ≥ 1.5 – 2 mm in preterm infants), but absolute size is not enough — you must also consider flow pattern and pressure gradient.
- Large ductal diameter + low velocity indicates minimal resistance to flow $\rightarrow$ most likely unrestrictive.

2. Velocity / Pressure Gradient

- Low peak systolic velocity (<2 m/s) on Doppler suggests that the pressure gradient between the pulmonary artery and aorta is small, so there is generally no major restriction.
- For bidirectional shunt:
 - Right-to-left systolic flow with velocity <2 m/s = usually no restriction.
 - Left-to-right diastolic flow with velocity <1.5 – 2 m/s = systemic–pulmonary diastolic gradient small $\rightarrow$ still unrestrictive.

3. Flow Profile (Continuous vs. Phasic)

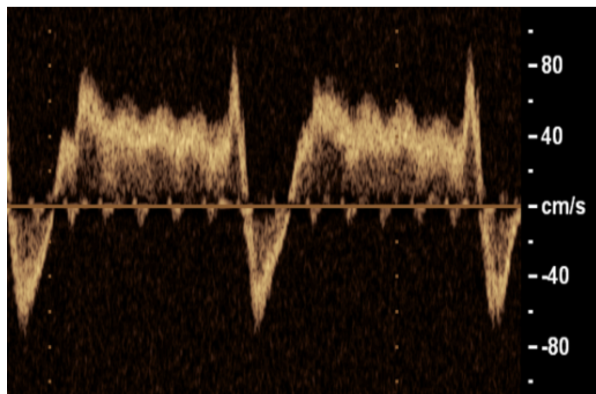
- Unrestrictive PDA shows a “plateau” Doppler shape, with broad components and minimal tapering.
- Restrictive ducts show “peaky,” high-velocity, short-duration signals (like a “fencing saber”) because of high gradient across the duct.

4. Aortic and Pulmonary Pressure Relationship

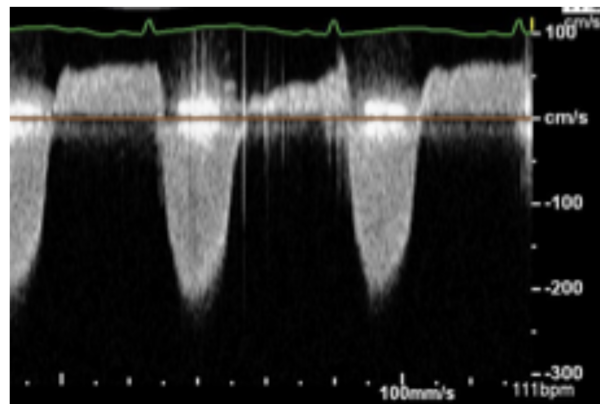
- If PDA is unrestrictive, PA systolic pressure $\approx$ systemic systolic pressure (equalization).
- This can be confirmed by TR gradient (for sPAP) or IVS position (mid-systolic septal flattening).
- When pressures have equilibrated, flow depends mainly on relative SVR vs. PVR $\rightarrow$ leads to bidirectionality (right-to-left in systole, left-to-right in diastole).

Bidirectional PDA that is unrestrictive. In this case, the velocities are low and there is rapid equalization of pressure on both end of the duct with low systolic Right to Left velocities (50 cm/s) and low diastolic velocities (60-75 cm/s).

Bidirectional PDA that is restrictive (subjective interpretation). There is a relatively higher systolic right to left component at 2 m/s, outlining that there is some pressure difference between the PA and the Ao in systole (at least 16 mmHg). The diastolic component is also more restrictive with a left to right shunt at 80 cm/s, outlining some restriction to pressure transmission.



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